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SIMPLE SUBSTITUTION CIPHER



The Simple Substitution Cipher substitutes one letter for another. Since there are 26 possible substitutions for the letter *A*, 25 possible substitutions for *B*, 24 for *C*, and so on, the total number of possible keys is $26 \times 25 \times 24 \times 23 \times \dots \times 1$, or 403,291,461,126,605,635,584,000,000 keys! That's far too many keys for even a supercomputer to brute force, so the code-breaking method used in Project 7, “Caesar Hacker,” can't be used against the simple cipher. Unfortunately, devious attackers can take advantage of known weakness to break the code. If you'd like to learn more about ciphers and code breaking, you can read my book *Cracking Codes with Python* (No Starch Press, 2018; <https://nostarch.com/crackingcodes/>).